A Search for Lorentz Invariance and CPT Violation using Neutrino Interactions at NOνA

Krishna Lakkaraju, Brian Rebel (Advisor)

* *E-Mail: Klakkaraju@wisc.edu*

Abstract

The Standard Model of Particle Physics relies on fundamental symmetries, including CPT symmetry and Lorentz invariance. CPT symmetry asserts that particle interactions remain the same under the combined operations of charge conjugation (C), parity inversion (P), and time reversal (T). Lorentz invariance requires that physical laws remain invariant under spatial and temporal transformations. Violation of either symmetry would indicate new physics beyond the Standard Model. To search for such violations, we will analyze neutrino interactions recorded by the NuMI Off-Axis ν_e Appearance (NOvA) experiment. We will measure variations in the neutrino interaction rate as a function of sidereal time, where potential violations of Lorentz invariance and CPT are predicted. This work proposes the first search for Lorentz-invariance driven oscillations between muon and electron neutrinos at NOνA. This analysis extends beyond previous disappearance studies by utilizing the high-statistics ν_e dataset.

1. Introduction

There are several fundamental principles of physics. For instance, the principle of Lorentz Invariance states that the laws of physics must be the same everywhere in the universe at all times. CPT symmetry is a fundamental principle that states that the laws of physics remain the same under successive operations of charge conjugation (C), parity inversion (P), and time reversal (T). Charge conjugation replaces particles with their antiparticles, parity inversion reflects the spatial coordinates, and time reversal inverts the direction of time evolution. Although its constituent symmetries may be violated, combined CPT symmetry violation has not yet been observed. The current formulation of the Standard Model of Particle Physics assumes that Lorentz invariance and CPT symmetry hold.

Given how fundamental the principles of Lorentz Invariance and CPT symmetry are to physics, we must test them experimentally. The Standard Model Extension (SME) is a model that unites quantum gravity with particle interactions. The SME includes coefficients which describe all possible violations of Lorentz invariance and CPT symmetry at the Planck scale, 10^{19} GeV [1].

Neutrinos provide a way to search for Lorentz Invariance violation (LV) and CPT violation (CPTV). The SME predicts that violations may drive transitions between neutrino flavors (ν_e, ν_μ, and ν_τ). If LV-driven oscillations occur, they will manifest as modulations in the neutrino interaction rate, dependent on the propagation direction relative to the Sun-centered inertial reference frame [1]. Each full rotation of the Earth, or sidereal day, has a duration of approximately 23 hours and 56 minutes, which is slightly shorter than a solar day [2]. Detecting a periodic variation in the neutrino interaction rate as a function of sidereal time would be evidence of LV and CPTV.

The NuMI Off-axis Neutrino Appearance (NOνA) experiment is pivotal to our proposed analysis. By utilizing the NuMI beam and near detector (ND) located 1 km from the neutrino source, NOνA provides a unique opportunity to investigate neutrinos for sidereal modulation. At this distance, any neutrino oscillations would be sensitive to symmetry-violating effects predicted by the SME. Previous searches for Lorentz violation using the Main Injector Neutrino Oscillation Search (MINOS) primarily analyzed the ν_μ disappearance channel, leaving LV-driven ν_e appearance largely untested. This analysis will be the first to search for LV and CPTV using the large NOνA ν_e dataset at a baseline of 1 km and energies of approximately 2 GeV. The inclusion of both a ν_e and a ν_μ dataset may permit greater sensitivity to small sidereal modulations as predicted by the SME.

2. Methods

The NOνA experiment utilizes a particle accelerator located at Fermilab to accelerate a beam of protons. The accelerated protons impinge on a graphite target to produce hadrons, namely kaons and pions [4]. When these hadrons decay, they produce neutrinos. The number of neutrinos produced is proportional to the quantity of protons that strike the target. The neutrinos generated

constitute the beam that travels toward the ND, which is shown in figure 1. We are interested in the rate at which neutrinos from the beam interact with ions in the ND. Since the NuMI beam primarily consists of muon-neutrinos, the ND is particularly sensitive to LV-driven oscillation signals in the ν_e channel. If LV and CPTV exist, we would observe periodic variations in the rate of ν_e interactions as a function of sidereal time.

The NuMI beam delivers approximately 5×10^{13} protons to the graphite target for a duration of $10\mu s$ every $1.7s$ [5]. Each proton delivery is tagged with a global positioning system (GPS) that records its time signature in Universal Coordinated Time (UTC) [5]. The GPS records when the protons are extracted into the main injector using a pulsed dipole magnet, ensuring the precise event timing required for this analysis [5]. We will collate the resulting timing data into a histogram of the number of neutrino interactions per unit of protons delivered to the target (POT), as a function of sidereal time. To reduce noise, we will use the standard NOνA data-quality criteria to select periods with acceptable beam performance.

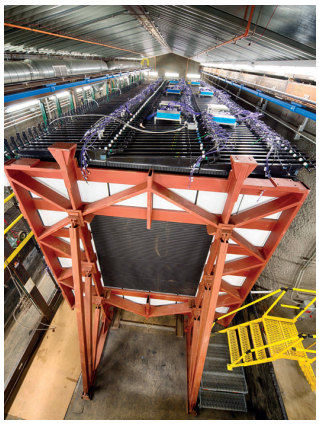

Figure 1. Image of the NOνA near detector. We analyze neutrinos from the NuMI beam after they interact with scintillator nuclei [3].

We will utilize a Fast Fourier Transform (FFT) to decompose the data into integer multiples of the sidereal frequency, $\omega_\oplus$, and corresponding harmonic signal powers. We will restrict our search to the first four sidereal harmonics, as predicted by first-order sidereal modulations from the SME theory. The harmonic powers output by the FFT quantify the amplitude of sidereal variation in the NOνA data. To interpret these harmonic powers, we will simulate many experiments with identical exposure to the data. Each simulated experiment will assign events from each spill a random sidereal time, yielding an event rate as a function of sidereal time containing zero sidereal

variation. Performing the FFT on each randomized distribution establishes the signal threshold, as shown in figure 2.

If the harmonic powers calculated from the NOνA data exceed the signal threshold, there is a significant level of sidereal variation, implying LV and CPTV effects. Conversely, FFT powers within the threshold do not provide statistically significant evidence of sidereal variations. In this case, we will perform additional simulations to constrain symmetry-violating coefficients from the SME [1] which have not yet been directly tested. We will place limits on these coefficients by increasing their value and performing an identical FFT analysis to determine the theoretical values necessary to exceed the signal detection threshold. We will consider LV-driven oscillations in and between the ν_μ and ν_e datasets. Since the NuMI beam is primarily comprised of ν_μ, small LV-driven $\nu_\mu \longrightarrow \nu_e$ signals would be identified in the ν_e channel.

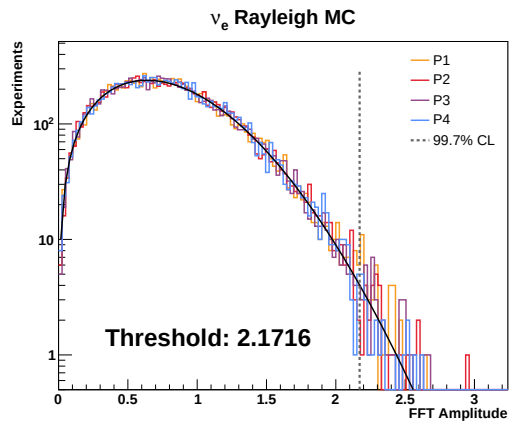

Figure 2. Distribution describing the possible spectrum of FFT powers for each of 10^4 simulated null experiments.

3. Timeline & Contributions

The remainder of the work will take place over seven months. In the first four months, we will finalize the signal detection thresholds and SME coefficient limits, while considering systematic effects on the data. The following three months will be used to prepare results for submission to *Physical Review D* for peer review. Through collaborative effort with a research advisor and the broader NOνA collaboration, I will lead implementation of the FFT, Monte Carlo threshold,

and coefficient sensitivity analyses. In addition, I will develop the simulation of LV-driven oscillations between ν_μ and ν_e to constrain SME coefficients which have not been tested in previous disappearance analyses at MINOS. During the development of the analysis, I will present progress updates and findings at NOνA collaboration meetings.

4. Conclusion & Future Directions

This work explores violations of fundamental symmetries critical to both experimental and theoretical particle physics. Detecting sidereal modulation would provide evidence for phenomena beyond the Standard Model, thereby developing a new sector for exploration within physics. Contrarily, the absence of sidereal variation would further constrain extensions to the Standard Model. Regardless of the results, this research will contribute to the broader effort to probe the fundamental symmetries that underlie modern particle physics. Future extensions of this work include extensions of the analysis to the NOνA far detector, considering oscillations between neutrinos and anti-neutrinos.

References

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